Project Title: LPS_14V_V0.1

A simple Linear Power Supply (LPS) designed to convert **24V AC** to a regulated **14V DC** output.

Technical Specifications

- **Input:** 24V AC (via external step-down transformer)
- **Output:** 14V DC Regulated
- **Regulator:** LD1086D2M33TR (Note: Ensure the feedback resistor values match your target 14V output)
- **Rectification:** Full-bridge rectifier (1N4007 diodes)
- **Filtering:** Dual electrolytic capacitor stage

Project Structure

- `/Hardware`: Contains KiCad schematic (`.kicad_sch`) and PCB layout (`.kicad_pcb`).
- `/Gerbers`: Production-ready files for PCB manufacturing.
- `/Docs`: Schematic PDF and 3D renders.

Technical Observations & Tips for your Design

Before you push the final version, you might want to double-check these areas in your KiCad files:

1. The Regulator Circuit (U1)

In your schematic, you are using the **LD1086D2M33TR**.

- **Note:** The "33" in the part number usually indicates a fixed **3.3V** output. If you intend to get **14V** using the feedback resistors (R1/R2), ensure you are using the **Adjustable** version of the LD1086.
- **Calculation:** The output voltage for an adjustable regulator is typically calculated as:

  $$V_{out} = V_{ref} \left(1 + \frac{R_2}{R_1}\right) + (I_{adj} \times R_2)$$

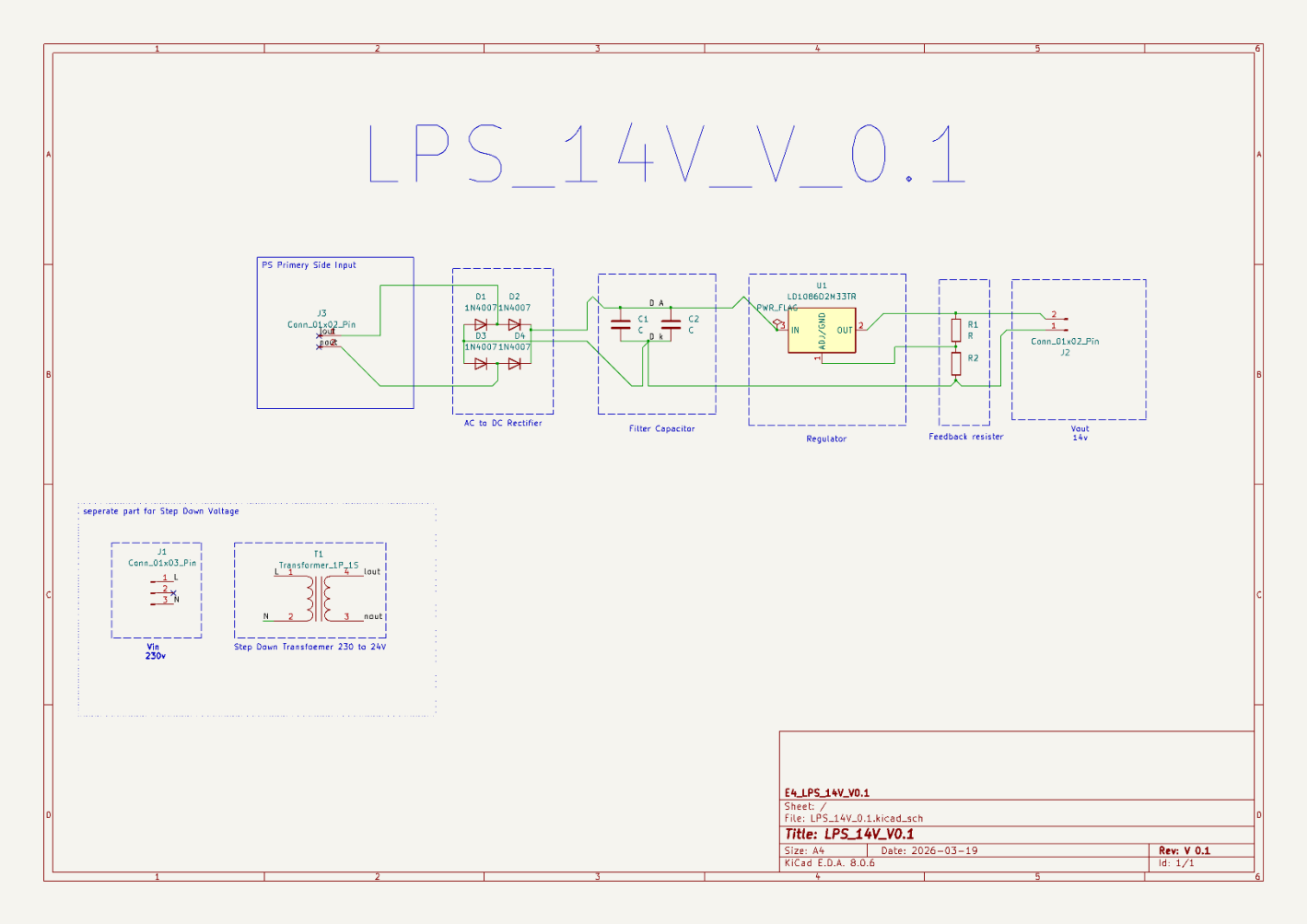
  (Where V_{ref} is usually 1.25V).

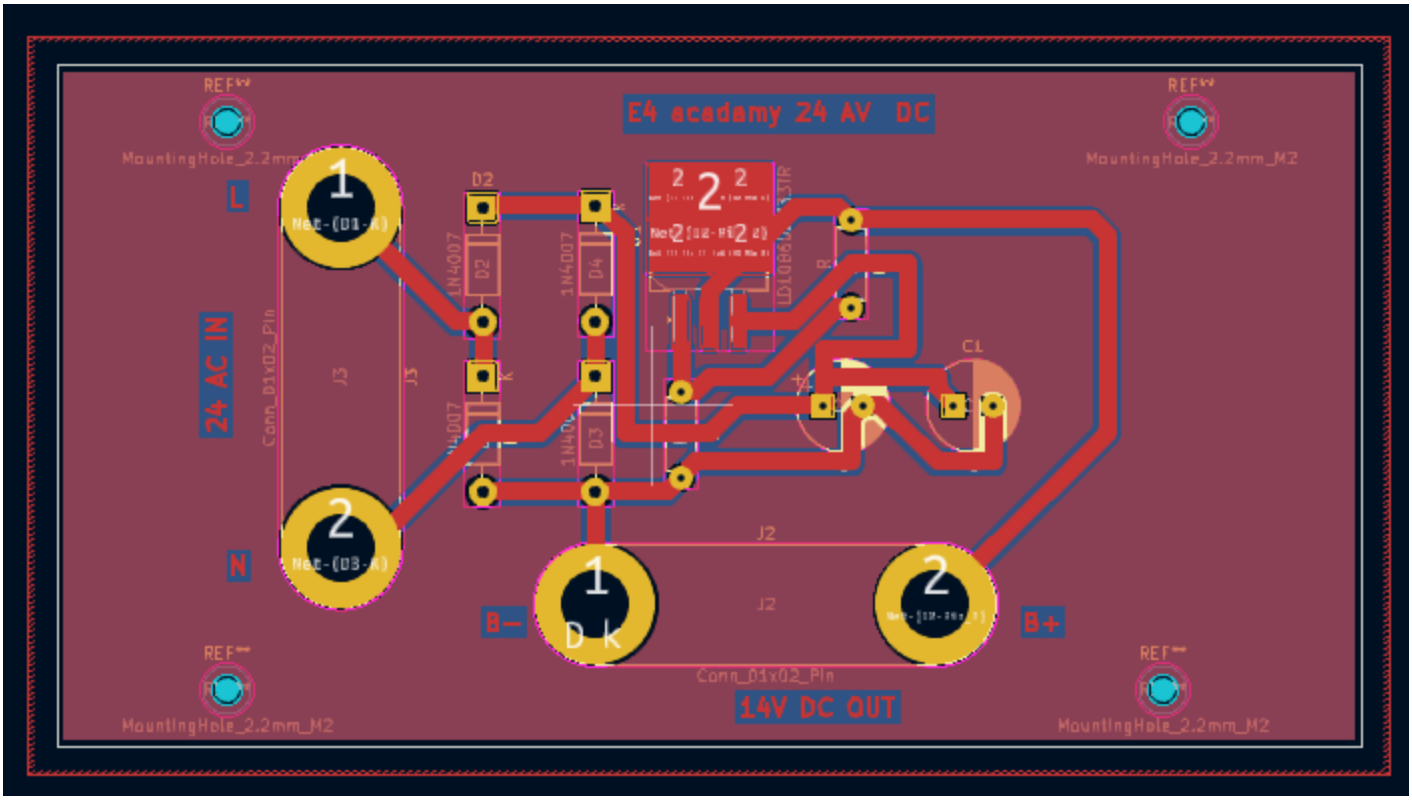
2. PCB Layout Improvements

- **Trace Width:** The traces from the AC input through the rectifier and filter capacitors carry the most current. Consider thickening these traces to reduce resistance and heat.
- **Thermal Relief:** Your 3D view shows the regulator (U1) is an SMD component. Since linear regulators dissipate heat based on the voltage drop ($Input - Output$), ensure you have a large copper pour (heatsink) connected to the tab of U1 on the PCB.
- **Clearance:** Ensure the clearance between your AC input pads and the ground plane meets safety standards for your specific application.

3. Repository "Pro-Tips"

- **Include a Bill of Materials (BOM):** Use KiCad's "Generate BOM" tool and export it as a `.csv` so others know exactly which parts to buy.
- **Interactive HTML BOM:** There is a great KiCad plugin called "InteractiveHtmlBom." Including its output in your GitHub repo makes it much easier for others to assemble the board.




E4 acadamy 24 AV DC
REF**
MountingHole_2.2mm_M2
L
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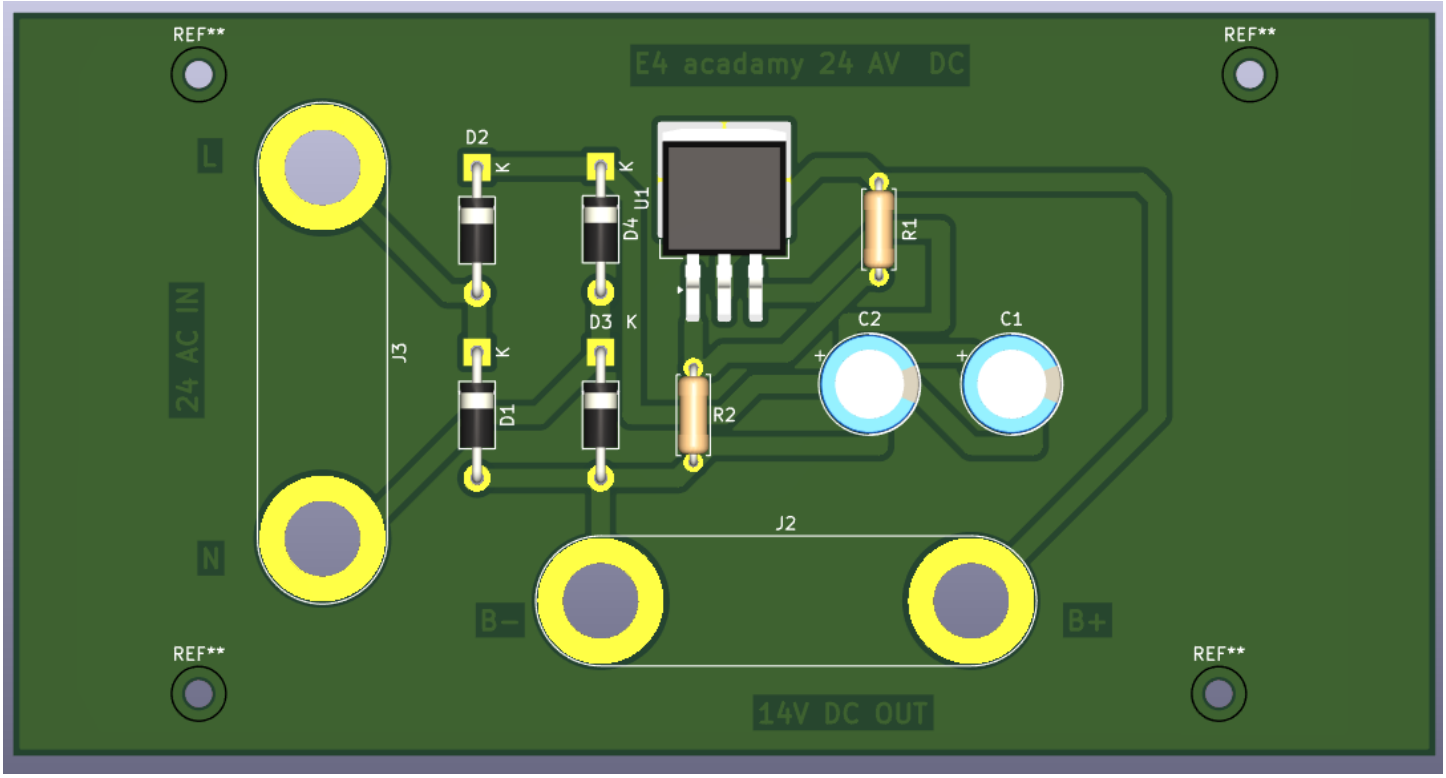
24 AC IN
J3
Conn_01x02_Pin
D2
D3
D4
1N4007
C1
J2
B−
B+
D K
14V DC OUT


E4 acadamy 24 AV DC
REF**
L
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24 AC IN
J3
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J2
B−
B+
14V DC OUT